Data Generation Machinery for N-of-1 Clinical Trial Simulations

pmsimstats team

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Contents

1 Abstract

Background. Simulation is the standard tool for power and sample-size calculation in aggregated N-of-1 trials. In this design each participant is repeatedly crossed on and off treatment, and many single-patient series are then pooled to validate a predictive biomarker. Any such simulation must specify how the biomarker moderates the treatment effect in its data-generating process (DGP). That specification is a substantive choice, not an implementation detail. Each of the two established parameterizations carries its own advantages and its own costs. We formalize both and ask one question. In the Hybrid N-of-1 design that motivates this software, why does carryover erode power so much more under one architecture than under the other?

Methods. We define two DGP architectures. Under 'mean moderation' the biomarker scales the treatment effect in the conditional mean. Under 'covariance moderation' the interaction is instead represented as a correlation that depends on drug-exposure condition (on drug versus off drug), within a joint multivariate normal (MVN) distribution. We hold the analysis model fixed at a linear mixed model with continuous-time AR(1) residual correlation. We then use Monte Carlo simulation to quantify the power to detect the interaction as carryover increases in the Hybrid N-of-1 design, the within-subject design used by the prazosin and PTSD reference application that motivates this software. We also run a traditional crossover (CO) and an open-label-plus-blinded-discontinuation design (OL+BDC) in parallel, at the same total N. These serve as contrast cases that isolate the design feature driving the architecture-dependent divergence.

Results. Under mean moderation, power is nearly preserved in the Hybrid design. Carryover does not disrupt the biomarker's relationship to the treatment effect here. At $N = 70$ power falls only from 0.842 to 0.784 as the carryover half-life grows to one week, a relative loss of 6.9% ($z = 3.34$). Under covariance moderation the same carryover instead erodes the differential correlation that carries the signal. Power falls from 0.840 to 0.711, a relative loss of 15.4% ($z = 2.80$, $p = 0.005$), more than double the mean-moderation loss. We trace this gap to a two-channel erosion mechanism. Carryover shrinks the drug-exposure contrast, a channel shared with mean moderation, and it independently decays the biomarker-response correlation itself, a channel unique to covariance moderation. The damage from this second channel scales with how much uninterrupted off-drug time a design exposes patients to. That is why the architecture contrast is negligible in the classical crossover (CO, $z = 0.29$), where within-subject AR(1) correlation already carries most of the signal covariance moderation depends on. The contrast is far larger in the open-label-plus-blinded-discontinuation design

(OL+BDC, $z = 7.64$, $p < 10^{-13}$). Its extended blinded-discontinuation phase gives the correlation-decay channel the most uninterrupted time to act. The Hybrid design alternates on- and off-drug periods rather than sustaining either for long, so its intermediate carryover exposure places its architecture-dependent power loss between these two extremes.

Keywords. N-of-1 trials; biomarker-treatment interaction; data-generating process; multivariate normal; carryover effects; clinical trial simulation; precision medicine; statistical power.

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2 1. Introduction

A crossover trial is built on balance: every participant moves through the same sequence of equal-length periods and receives each treatment once. We begin with the three designs compared here, which span the departure from that balance. At one end sits a conventional two-period crossover. At the other sits a hybrid design combining an unblinded open-label phase, a blinded randomized-withdrawal phase, and a single crossover cycle. Measurement spacing, randomization, and blinding all differ from phase to phase across this span.

N-of-1 and aggregated N-of-1 designs are being adopted with increasing frequency across clinical and behavioral research[1,2]. Systematic and scoping reviews document this growth across fields as varied as pediatric psychiatry[3], neurology[4], epilepsy care[5], palliative care[6], translational nutrition[7], and the single-case experimental designs of psychology and education[8]. A scoping review now assesses the effect of the design on clinical outcomes themselves[9], and the design has acquired both a CONSORT reporting extension and a national policy agenda[10,11].

The principal advantage of these designs over a parallel-group alternative is efficiency. Because each participant serves as their own control, the between-participant variance component is removed from the treatment contrast. Pooling a modest number of individual series in a mixed model therefore yields population-level inference at sample sizes far below those a parallel-group trial would require[12–14]. This efficiency gain has been quantified directly, in simulation-based comparisons of aggregated N-of-1 trials against parallel and crossover designs[15,16] and in a developed literature on sample-size and power calculation for the N-of-1 setting, including the serial correlation that repeated within-participant measurement induces[17–20].

Increasingly the target of inference in these designs is not the average treatment effect but a biomarker-treatment interaction. That is, the question is whether a baseline marker identifies which participants respond[21–24]. We note that the repeated within-participant measurement that makes N-of-1 designs efficient for main effects also makes them particularly attractive for this purpose, since interaction tests are notoriously underpowered in parallel-group trials. Aggregated N-of-1 designs have accordingly been proposed and evaluated specifically as a vehicle for predictive-biomarker validation[25–27].

Establishing such an interaction requires knowing the power of the test for it. For simplified special cases, such as a balanced strict repeated-measures analysis of variance under sphericity, that power does have a closed form, and a companion paper[28] derives it together with its equivalence to a two-sample *t*-test. For the continuous-time mixed model and the N-of-1 designs considered here, however, no such closed form is available. Power is instead estimated by Monte Carlo simulation: many trials are generated and analyzed, and the rejection rate is recorded. The simulation must itself specify the biomarker-treatment interaction in its data-generating process, and, as we shall show, there is more than one way to do so.

We distinguish two architectures. The common one, direct mean moderation, adds an interaction term to the outcome equation so that the biomarker scales the size of the treatment effect. Essentially every standard trial-simulation framework uses it[22,23]. The second, covariance moderation through differential correlation, is the specification Hendrickson et al.[25] used for N-of-1 trials. It instead makes the biomarker change the correlation between treatment response and outcome, so that the interaction is represented in the joint distribution rather than in the mean.

On this reading, covariance moderation is the reduced-form second-moment representation of a latent responder-class mechanism, a point developed at length in a companion paper[29]. Patients may belong to unobserved responder and non-responder groups, with the biomarker only an imperfect indicator of that status, so that two patients with the same biomarker value can still respond differently. Mean moderation cannot represent this situation, because it assumes that the biomarker fixes each patient's drug effect exactly. Covariance moderation represents it instead by letting biomarker and response track one another only while the drug is acting, as a tendency rather than a fixed rule, and that tendency fades once the drug washes out. As Section 4 shows, that difference is why carryover costs more power under covariance moderation than under mean moderation.

A third, 'combined' architecture activates both channels simultaneously through two independent parameters and recovers these two as its single-channel limits. Because it introduces a second free parameter, and because its power behavior warrants separate treatment, we develop it in a companion paper[30] and do not consider it further here.

Choosing between these two architectures is not a matter of computational convenience, because each carries a genuine advantage and a genuine cost. On the one hand, mean moderation is simple, has a single free parameter, and preserves the biomarker-treatment effect ratio under carryover. It commits, however, to a strong assumption, namely that the biomarker deterministically fixes each patient's drug response. That assumption is implausible whenever the biomarker is only a statistical proxy for an unobserved responder subtype.

Covariance moderation, on the other hand, relaxes that assumption. As the reduced-form representation of a latent responder-class mechanism discussed above, it lets the biomarker predict response only probabilistically. This is the more defensible choice whenever responder status, rather than the biomarker itself, is the true causal driver, as is plausibly the case for the noradrenergic-tone hypothesis behind the prazosin and PTSD reference application used throughout this paper. That fidelity to the underlying biology comes at a real cost. Because the covariance-moderation signal lives in a correlation between biomarker and response that is sustained only while the drug is active, any off-drug erosion of drug effect directly erodes the statistical signal itself. This paper shows that the resulting power losses can run many times larger than under mean moderation. Investigators who adopt covariance moderation on biological grounds should therefore budget for its carryover cost at the design stage, rather than discover it after a trial has already been powered.

Hendrickson et al.[25] chose covariance moderation for exactly this reason when they designed the Monte Carlo simulation program behind the prazosin and PTSD trial. Given that trial's own results, covariance moderation was the logical, biologically appropriate specification. Representing the interaction in the mean would have imposed a deterministic biomarker-response relationship that the trial's biology did not support. Mean moderation, by contrast, was and remains the modal specification across the N-of-1 and broader biomarker-simulation literature (Section 4.4), so Hendrickson's choice was a deliberate departure from convention rather than an oversight.

What that departure costs, measured in statistical power under carryover, had not previously been quantified. Neither had the mechanism producing that cost been traced to specific, design-relevant features of the trial. Doing both is the central aim of this paper. We take the covariance-moderation choice as scientifically justified for the prazosin and PTSD application. In the Hybrid design that trial and this software's target applications use, we ask why carryover erodes power so much more severely under covariance moderation than under mean moderation, so that investigators reasoning by analogy from that trial can budget for the penalty explicitly and understand which design features control its size.

Within-participant designs are attractive for this question because of precision, not because they resolve individual response. Interaction parameters are estimated less precisely than main effects, for a binary marker by a factor equal to the inverse of the marker's prevalence. The advantage of a within-participant over a parallel-group design for that interaction depends on the complement of the intraclass correlation[31]. We note that this result is derived for a two-period crossover with a binary marker, so it orients rather than describes the present setting, in which the biomarker is continuous and the designs run over many timepoints. Separating a genuine participant-by-treatment variance component from within-participant error is a different problem. It requires repeated treatment cycles within each participant[14,17,32], and none of the designs studied here provides that. The estimand throughout is accordingly the fixed biomarker-by-exposure coefficient of Section 2.1, and the latent-responder reading of residual heterogeneity is developed in a companion paper[29].

The feature that distinguishes these designs analytically is *carryover*: residual drug effect that decays over days to weeks after discontinuation and contaminates off-drug occasions. Our focus is the Hybrid N-of-1 design, the within-subject design behind the prazosin and PTSD reference application and the design this software was built to support. It alternates on-drug and off-drug periods rather than sustaining either for long. We run two further within-subject designs in parallel as contrast cases, chosen to isolate which feature of the Hybrid design's carryover exposure drives its architecture-dependent power loss. The first is a traditional crossover (CO), whose within-subject AR(1) correlation dominates any covariance-moderation signal. The second is an open-label-plus-blinded-discontinuation design (OL+BDC), whose extended blinded phase gives carryover the most uninterrupted time to act of the three. All three designs are defined in Section 3.

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3 2. Methods

3.1 2.1 Notation and analysis model

Throughout, B_i denotes the biomarker value of participant i, D_{it} the binary on-drug indicator at timepoint t (1 on drug, 0 off drug), and BR_{it} the biological response component of the outcome. These three symbols describe the data-generating-process layer. The analysis-model layer uses a related but distinct continuous drug-exposure variable, `Dbc`, the continuous analogue of D_{it}. It equals 1 during on-drug periods and $\exp(-\lambda t_{sd})$ during off-drug periods, where t_{sd} is time since drug discontinuation and λ is the carryover decay rate. This rate is typically derived from the drug's pharmacokinetic half-life as $\lambda = \ln 2/t_{1/2}$. We note that whereas D_{it} flips abruptly between 0 and 1, `Dbc` captures the smooth shrinkage of residual drug effect during the off-drug period. Code-style identifiers in typewriter font refer to variables in the pmsimstats codebase or to R formula syntax; mathematical symbols are used otherwise. The analysis model is the same under both architectures: an `nlme::lme` fit with a `corCAR1` continuous-time autocorrelation structure, a random intercept, and a single biomarker-by-drug interaction term, written `bm:Dbc` in R formula syntax. The interaction coefficient on `bm:Dbc` tests whether the biomarker moderates the exposure-decayed drug effect.

3.2 2.2 Two data-generating architectures

3.2.1 2.2.1 Mean moderation

In this architecture the biomarker enters the mean structure of the outcome directly. The treatment effect for participant i at timepoint t is

$$Y_{it} = \mu_0 + \beta_1 \cdot D_{it} \cdot (1 + \beta_{bm} \cdot B_i) + \epsilon_{it},$$

where D_{it} is the on-drug indicator (or a continuous measure of drug exposure), B_i is the participant's biomarker value (typically centered), and β_{bm} is the biomarker moderation parameter. The interaction is explicit, deterministic, and lives in the first moment of the outcome. Participants with higher biomarker values have proportionally larger treatment effects, and this population-level ratio is preserved under any proportional scaling of D_{it}, including the shrinkage that carryover imposes.

This multiplicative form maps directly onto the standard additive regression form for predictive biomarker effects used in the trial methodology literature, specifically the effect-modeling equation of Kent et al.[21], $Y_{it} = \beta_0 + \beta_1 D_{it} + \beta_2 B_i + \beta_3 D_{it} B_i + \epsilon_{it}$, with $\beta_3 = \beta_1 \cdot \beta_{bm}$. The additive form additionally accommodates a prognostic main effect β_2 for the biomarker, which the multiplicative form sets to zero by construction. For power calculations on the interaction coefficient, however, the two forms are interchangeable up to this prognostic-effect distinction, and both fall within mean moderation.

The implementation realizes this architecture in the equivalent conditional-mean parameterization rather than in the multiplicative form of the display above. At each on-drug occasion the response component of participant i is shifted by

$$\Delta_i = c_{bm} \cdot \frac{\sigma_{BR}}{\sigma_B}(B_i - \mu_B) = c_{bm}\, z_i\, \sigma_{BR},$$

where $z_i = (B_i - \mu_B)/\sigma_B$ is the standardized biomarker, and by nothing at all at off-drug occasions. Comparing this with the multiplicative display, the two coincide when $\beta_1\beta_{bm} = c_{bm}\sigma_{BR}/\sigma_B$, so a single parameter suffices and c_{bm} is used for both architectures throughout.

That shift is, by construction, the on-drug conditional mean that covariance moderation induces through the covariance, given at the end of Section 2.2.2. The two architectures therefore agree exactly in the first moment while the drug is acting, at equal c_{bm}, and differ only in whether the biomarker-response association is written into the second moment as well. This is what licenses running both at a common parameter value: the power differences reported in Section 3 are differences in where the interaction is encoded, not differences in the magnitude of the on-drug effect.

One consequence of this gating deserves stating explicitly, because it is a point on which the data-generating process and the analysis model do not agree. The shift is applied at on-drug occasions and nowhere else, so the mean-moderation DGP places no interaction signal at off-drug occasions, whatever the carryover half-life. The analysis model of Section 2.1, however, regresses on the continuous exposure variable `Dbc`, which equals $e^{-\lambda t_{sd}}$ and is therefore nonzero at off-drug occasions whenever carryover is present. The fitted `bm:Dbc` coefficient consequently looks for a graded off-drug interaction that this architecture does not generate, whereas under covariance moderation the two use the same exponential form and agree.

The assumption embedded in mean moderation is therefore that a biomarker which moderates the drug effect ceases to moderate anything once administration stops, even while residual drug effect persists in the response mean. That is defensible, since the biomarker scales an administered treatment effect rather than a pharmacological concentration, but it is an assumption rather than a neutral implementation choice, and it contributes to the architectures' differing sensitivity to carryover reported in Section 3.

3.2.2 2.2.2 Covariance moderation through differential correlation

In this architecture the biomarker and the biological response (BR) component are jointly drawn from a multivariate normal distribution with a correlation that depends on drug-exposure condition,

$$\begin{pmatrix} B_i \\ BR_{it} \end{pmatrix} \sim \text{MVN}\left(\begin{pmatrix} \mu_B \\ \mu_{BR}(t, D_{it}) \end{pmatrix}, \Sigma(D_{it})\right), \qquad \text{Cor}(B_i, BR_{it}) = \begin{cases} c_{bm} & D_{it} = 1 \\ c_{bm} \cdot e^{-\lambda t_{sd}} & D_{it} = 0, \end{cases}$$

with the exponential-decay form as the default. The limiting cases $\lambda = 0$ (no decay) and $\lambda \to \infty$ (immediate decay) recover the two no-carryover model variants. The population mean of BR_{it} is the same for all participants with the same treatment history, so the interaction is not in the mean structure. It instead emerges from the conditional distribution. Given $B_i = b$,

$$E[BR_{it} \mid B_i = b, D_{it} = 1] = \mu_{BR}(t) + c_{bm} \cdot \frac{\sigma_{BR}}{\sigma_B} \cdot (b - \mu_B).$$

The interaction here is probabilistic rather than deterministic, lives in the second moment of the joint distribution, and persists with attenuated strength off drug, vanishing only in the limit of complete decay. Carryover in the DGP inflates the BR means during off-drug periods, making them resemble on-drug means. This reduces the differential in the correlation structure between drug-exposure conditions. Section 2.2.3 shows that this erosion leaves the analysis model's point estimate of the interaction largely unbiased, but it converts variance the biomarker previously explained into unexplained residual noise, which is the channel through which it costs power (Section 3.2). A third, combined architecture, in which both channels operate at once through independent weights, recovers the two architectures above as its single-channel limits and is the natural framework when the operative mechanism is uncertain. It is developed in a companion paper[30].

3.2.3 2.2.3 Analytical comparison

Consider the expected outcome difference between two participants with biomarker values b_1 and b_2 at the same drug exposure level D. Under mean moderation,

$$E[Y \mid b_1, D] - E[Y \mid b_2, D] = \beta_1 \cdot \beta_{bm} \cdot (b_1 - b_2) \cdot D.$$

This scales linearly with D and therefore shrinks during off-drug periods, when D is replaced by an exposure-decayed value. However, the *ratio* of drug effect between high- and low-biomarker participants is preserved at every level of exposure, so the biomarker-treatment signal contracts in amplitude but does not lose identifiability under carryover. Under covariance moderation the analogous quantity, during off-drug periods with carryover, is

$$E[BR \mid b, D = 0, t_{sd}] = \mu_{BR,\text{off}}(t_{sd}) + c_{bm} \cdot e^{-\lambda t_{sd}} \cdot \frac{\sigma_{BR}}{\sigma_B} \cdot (b - \mu_B).$$

Here the biomarker-dependent component decays exponentially with time off drug. As t_{sd} increases, the off-drug conditional expectation converges to $\mu_{BR,\text{off}}$ for all biomarker values, in contrast to mean moderation's preserved ratio.

This convergence in the raw time since discontinuation is not, on its own, the source of the power loss documented in Section 3, and disentangling the two requires re-expressing the comparison in terms of the regressor the analysis model actually uses. Write the exponential decay factor as $\kappa_{it} = 1$ on drug and $\kappa_{it} = e^{-\lambda t_{sd,it}}$ off drug, matching the definition of $D_{bc,it}$ in Section 2.1 exactly when the analysis-model half-life is set equal to the DGP half-life, the base case examined in Section 3.1. Then $\mathrm{Cor}(B_i, BR_{it}) = c_{bm} \cdot D_{bc,it}$ by construction, and the conditional mean above is *exactly linear* in $D_{bc,it} \cdot (b - \mu_B)$ at every value of t_{sd}, not only in the untouched limit:

$$E[BR_{it} \mid B_i = b, D_{it}] = \mu_{BR}(t) + c_{bm}\, D_{bc,it}\, \frac{\sigma_{BR}}{\sigma_B}(b - \mu_B).$$

Because the analysis model regresses the outcome on `bm:Dbc` using this same $D_{bc,it}$, its population (large-sample) estimate of the interaction slope recovers $c_{bm}\, \sigma_{BR}/\sigma_B$ *regardless of how much carryover has occurred*, to the same leading order that mean moderation's slope $\beta_1 \beta_{bm}$ is carryover-invariant. The biomarker-dependent signal is not eliminated by carryover in the sense that matters for the fitted interaction coefficient; only its raw amplitude at long t_{sd} is. This is confirmed empirically in Section 3.4: the mean of $\hat{\beta}_{bm:D_{bc}}$ under covariance moderation holds near -0.234 across every carryover level examined, rather than trending toward zero, exactly as this identity predicts.

If the population slope is unaffected, the power loss under covariance moderation must arise instead from the precision with which that slope is estimated, and the bivariate normal structure of Section 2.2.2 gives that precision loss in closed form. Conditional on B_i, the variance of BR_{it} is

$$\mathrm{Var}(BR_{it} \mid B_i, D_{it}) = \sigma_{BR}^2(1 - c_{bm}^2\, D_{bc,it}^2),$$

an exact consequence of the bivariate normal conditional-variance identity applied to a correlation of $c_{bm}\, D_{bc,it}$. On drug ($D_{bc,it} = 1$), the biomarker explains a fixed c_{bm}^2 fraction of BR's variance; as carryover erodes $D_{bc,it}$ toward zero off drug, that explained fraction shrinks *quadratically*, not linearly, in $D_{bc,it}$, and the corresponding conditional variance rises toward the full unconditional σ_{BR}^2, adding pure noise that the fixed-effects analysis model, which assumes homoscedastic residual variance, cannot distinguish from measurement error. Mean moderation has no analogue of this channel: its residual variance ϵ_{it} is constant by construction (Section 2.2.1), independent of D_{it}, so carryover there degrades only the regressor's contrast, never the noise floor around it. This is the quantitative content behind the qualitative two-channel account of Section 3.2: mean moderation loses information only through the design contrast in $D_{bc,it}$ itself, a channel shared by both architectures, while covariance moderation loses information through this quadratic variance-inflation channel as well, and it is this second, architecture-specific channel that produces the divergence in power documented in Section 3.

3.2.4 2.2.4 Relationship to Hendrickson et al.'s published implementation

Covariance moderation, as formalized in Section 2.2.2, develops Hendrickson et al.'s approach rather than reproducing it. We compared against their public repository (`github.com/rchendrickson/pmsimstats`) at commit `58b32a9` of 19 November 2020, the last commit preceding publication and therefore the code behind the published results. Commits made to that repository in 2024 alter the data-generating process substantially, and the comparison below is against the published state only. Four differences matter.

First, the temporal correlation structure. The published code assigns a single constant to every pair of timepoints within a factor, irrespective of their separation, which is compound symmetry. The architectures examined here instead use AR(1) decay, $\mathrm{Cor}(X_{t_i}, X_{t_j}) = \rho^{|t_i - t_j|}$, adopted for clinical plausibility, since nearby measurements should be more strongly correlated than distant ones, and for the numerical benefit described below.

Second, and most consequentially, the biomarker-response correlation. Both the published code and Architecture B make $\mathrm{Cor}(BM, BR_t)$ depend on drug state, but they do so through different gates and with different off-drug profiles. Architecture B gates on the drug indicator itself, taking the value c_{bm} at on-drug timepoints and $c_{bm} \exp(-\lambda t_{sd})$ off drug, so the correlation decays continuously with time since discontinuation. The published code instead gates on whether the response mean is nonzero, taking c_{bm} wherever it is and zero elsewhere, which is a step rather than a graded decay.

That difference in gate has a consequence that is easy to miss. Because carryover keeps the response mean nonzero after discontinuation, the position of the published code's step is itself carryover-dependent. At $t_{1/2} = 0$ the step coincides with the drug indicator and the two mechanisms are identical, timepoint for timepoint. As carryover increases they diverge in opposite directions: Architecture B's off-drug correlation decays toward zero, whereas the published code's rises to the full c_{bm}. At $t_{1/2} = 1.0$ in the Hybrid design the published code assigns c_{bm} at every timepoint, on drug and off, so its interaction channel carries no on-drug versus off-drug contrast at all. The graded off-drug decay that defines covariance moderation in Section 2.2.2 is therefore not a feature of the published implementation; it was introduced into that repository in 2024. Architecture B as studied here is accordingly developed from Hendrickson et al.'s stated rationale, that the biomarker is informative because it shares variance with drug-responsive physiology, rather than recovered from their published code.

Third, carryover. The published code does apply carryover, to the response mean only, adding $\mathrm{BR}_{p-1}(1/2)^{t_{sd}/t_{1/2}}$ to the response mean at off-drug timepoints. This paper retains that mechanism unchanged: the line implementing it in the current package is character-for-character the published one. The 2024 revision inserts an additional undocumented constant multiplying t_{sd} in that exponent, defaulting to two and thereby halving the effective carryover half-life. Because it postdates publication and carries no stated pharmacokinetic rationale, it is not used here.

Fourth, the analysis model. The published code fits `lmer` with a random intercept and no residual correlation structure, in which the exposure variable `Db` is a binary on-drug indicator. Its formula branches on whether expectancy varies within the design: for the Hybrid design, whose open-label phase is unblinded, the fitted model is `Sx ~ bm + De + Db + t + bm*Db + (1|ptID)`, carrying the expectancy term `De`; for the CO and OL+BDC designs, whose expectancy is constant, `De` is dropped. We use `nlme::lme` with `corCAR1` and the continuous exposure variable `Dbc` of Section 2.1, because a model that ignores residual autocorrelation under an autocorrelated data-generating process is misspecified in a way already known to distort inference, a defect independent of the architecture question this paper addresses.

The published source contains one further discrepancy, which we record because the repository's own history describes it as a fix and a reader may wonder whether it affected the published results. It did not. In the block assigning the cross-factor, same-timepoint correlation c_{cf1t}, the assignment `correlations[n1,n2]` appears twice, where the second was evidently meant to write the mirrored cell `correlations[n2,n1]`. That mirrored write is redundant here: the enclosing loop visits every *ordered* pair of factors, so the mirrored cell is assigned on a later pass regardless. We confirmed by direct computation that the published loop, the corrected loop, and a variant with the duplicated line removed all produce identical correlation matrices. The published correlation matrix is symmetric and correct, and this discrepancy is excluded from the four differences above.

The compound-symmetry-to-AR(1) change has a direct numerical consequence for the range of c_{bm} the simulations can explore. Compound symmetry restricts how large a biomarker-response correlation the covariance matrix can support before it ceases to be positive definite, because a constant correlation applied uniformly across many timepoints depletes the matrix faster than a correlation that decays with lag. [**TO RE-VERIFY: the specific positive-definiteness thresholds previously reported here were computed under a construction that combined compound symmetry with the decaying biomarker-response correlation, which is not the published configuration. Re-run against `58b32a9` before restoring numeric thresholds.**]

[**TO RE-RUN: the quantitative comparison arm previously reported in this subsection was executed against the 2024 state of the repository, including the scale-factor constant, and its power figures therefore do not describe the published implementation. Re-run the arm against `58b32a9` across the three designs and carryover half-lives of Section 3.1, and regenerate the accompanying heatmap, before restoring quantitative claims or the figure.**]

3.3 2.3 Simulation design

We ran identical simulation configurations under both architectures using the pmsimstats-ng framework, matching the DGP parameters as closely as possible. The non-biologic response components (time-varying response, placebo-belief accumulation, and residual noise) are generated from the same multivariate normal draw, with identical means, variances, within-factor AR(1) autocorrelation, and cross-factor correlations, under both architectures. We note that the two DGPs differ only in how the biomarker-to-biological-response linkage is specified. The divergent carryover behavior reported in Section 3 is therefore attributable to that parameterization alone, and not to any difference in how the other components are generated.

We follow the ADEMP framework of Morris, White, and Crowther[33], comprising Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures. The Aim is to compare power to detect the biomarker-by-drug interaction across architectures as carryover increases. The Data-generating mechanisms are the two architectures of Section 2.2. The **primary estimand** is the interaction coefficient $\beta_{bm:D}$ in the model `Sx ~ bm + t + Dbc + bm:Dbc`. The **Methods** setup is as follows.

- Trial designs: CO (traditional crossover), Hybrid (N-of-1 Hybrid), OL+BDC (open-label with blinded discontinuation).
- Sample size: $N = 70$ in total for every design, allocated across each design's randomization paths, so the three designs are matched on N. The two-path CO and OL+BDC designs put 35 on each path, and the four-path Hybrid design splits 70 as 18/18/17/17.
- Biomarker parameter: $c_{bm} = 0.45$ under both architectures, entering as the on-drug biomarker-response correlation under covariance moderation and as the standardized moderation coefficient of Section 2.2.1 under mean moderation. The two are calibrated to the same on-drug conditional mean, so this is a matched setting rather than a coincidence of labels.
- DGP carryover half-life: $t_{1/2} \in \{0, 0.5, 1.0\}$ weeks, with the analysis-model `Dbc` half-life matched to it.
- Replicates: 1,000 per cell (Monte Carlo standard error, MCSE, on a binomial power estimate near 0.5 is approximately 0.016).
- Implementation: 8-core `parallel::mclapply`, full factorial of 2 architectures $\times$ 3 designs $\times$ 3 $t_{1/2}$ $\times$ 2 c_{bm} levels gives 36 cells and 36,000 fits, with 100% convergence.

The **Performance measures** are power for $\beta_{bm:D}$ at $\alpha = 0.05$, Type I error under $c_{bm} = 0$ (both with MCSE $\sqrt{\hat{\pi}(1-\hat{\pi})/n_{\text{reps}}}$), bias of $\hat{\beta}_{bm:D}$ relative to the reference true value $\theta_{\text{true}} = -c_{bm} \cdot \sigma_{BR}$, the same calibration used in the companion carryover-specification and component-decomposition manuscripts[34,35], the empirical MCSE of $\hat{\beta}_{bm:D}$ across replicates, and the convergence rate. The number of replicates per cell is chosen to achieve MCSE below 0.02 for power at $\hat{\pi} = 0.5$, which corresponds to $n_{\text{reps}} \geq 625$. We use $n_{\text{reps}} = 1000$ throughout.

The $N = 70$ choice is the smallest of the prazosin-PTSD-matched sample sizes used in the published methodological literature. It is a regime in which no design or architecture saturates power at the no-carryover cells (the highest cell in the grid is 0.842), so carryover-induced erosion is mechanically visible. This is the main practical gain from matching the designs on N. We note that at a previous per-path parameterization the Hybrid cells sat above 0.99, where no erosion could register.

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4 3. Results

Covariance moderation, as Section 1 argues, is often the more biologically defensible choice when a biomarker is best understood as predictive rather than causal, and it is the specification Hendrickson et al. selected for the prazosin and PTSD application on exactly these grounds. That defensibility, however, does not eliminate a carryover cost. What follows quantifies that cost in the Hybrid design and traces why it arises, so that the appeal of the architecture does not obscure a real design consideration. The crossover (CO) and open-label-plus-blinded-discontinuation (OL+BDC) designs are reported alongside Hybrid at every step. They are not objects of primary interest in their own right; rather, contrasting Hybrid against these two structurally different designs is what lets us identify which feature of a design's carryover exposure drives the architecture-dependent loss.

4.1 3.1 Power under carryover at $N = 70$

Mean moderation.

Design	$t_{1/2} = 0$	$t_{1/2} = 0.5$	$t_{1/2} = 1.0$
CO	0.739	0.739	0.725
Hybrid	0.842	0.844	0.784
OL+BDC	0.751	0.748	0.730

Power declines only modestly under mean moderation. In the Hybrid design, the relative loss from $t_{1/2} = 0$ to $t_{1/2} = 1$ is 6.9%, and it is the only one of the three designs whose loss is separable from zero at 1{,}000 replicates per cell ($z = 3.34$). The two-period designs lose less (1.9%, CO; 2.8%, OL+BDC), and neither loss is distinguishable from Monte Carlo noise (CO $z = 0.71$; OL+BDC $z = 1.07$). All three values fall at or below the 8-13% relative-loss range suggested by earlier exploratory runs in this project for direct-mean-moderation DGPs. That is, this erosion leaves power substantially intact, even in the Hybrid design, where it is largest.

Covariance moderation (differential correlation, MVN).

Design	$t_{1/2} = 0$	$t_{1/2} = 0.5$	$t_{1/2} = 1.0$
CO	0.745	0.754	0.723
Hybrid	0.840	0.795	0.711
OL+BDC	0.777	0.694	0.539

Power declines more sharply under covariance moderation than under mean moderation in every design, but by very different amounts. In the Hybrid design, the relative loss from $t_{1/2} = 0$ to $t_{1/2} = 1$ is 15.4%, more than double its matched mean-moderation loss (6.9%). The two-period designs bracket this value from opposite sides. CO loses only 3.0%, not significantly different from mean moderation's CO loss, while OL+BDC loses **30.6%**, roughly 11 times its matched mean-moderation loss (2.8%) and twice the Hybrid loss. The CO cell shows a small, non-monotone fluctuation ($0.745 \to 0.754 \to 0.723$), within one MCSE and consistent with no architecture-by-carryover interaction in that design. These figures fall below the 40-60% relative-loss range suggested by earlier exploratory runs for the MVN differential-correlation DGP, but they confirm the qualitative finding of substantial, design-dependent erosion. We note that the Hybrid design occupies a genuine middle position here rather than resembling either contrast case.

We compare the carryover-induced power loss between architectures using a difference-of-differences z-statistic. Let $\Delta_X = \hat{\pi}_{X,0} - \hat{\pi}_{X,1}$ denote the within-architecture loss from $t_{1/2} = 0$ to $t_{1/2} = 1.0$ for architecture X, estimated from $n = 1{,}000$ independent replicates per cell. The covariance-minus-mean gap $\Delta_B - \Delta_A$ has standard error

$$\text{SE} = \sqrt{\frac{\hat{\pi}_{B,0}(1-\hat{\pi}_{B,0})}{n} + \frac{\hat{\pi}_{B,1}(1-\hat{\pi}_{B,1})}{n} + \frac{\hat{\pi}_{A,0}(1-\hat{\pi}_{A,0})}{n} + \frac{\hat{\pi}_{A,1}(1-\hat{\pi}_{A,1})}{n}},$$

since the four proportions come from independent replicate sets (all tests two-sided). In the Hybrid design, $\Delta_A = 0.842 - 0.784 = 0.058$ and $\Delta_B = 0.840 - 0.711 = 0.129$, giving $\Delta_B - \Delta_A = 0.071$ (about 7 percentage points) and $z = 2.80$ ($p = 0.005$). That is, the architectures' carryover behavior differs well beyond Monte Carlo error, even in this intermediate design. The same contrast is much larger for OL+BDC ($\Delta_A = 0.021$, $\Delta_B = 0.238$, $\Delta_B - \Delta_A = 0.217$, about 22 percentage points, $z = 7.64$, $p < 10^{-13}$), and it is indistinguishable from zero for CO ($z = 0.29$). This is because the CO design's within-subject AR(1) structure dominates the carryover-mediated correlation-decay channel that covariance moderation uses to represent the interaction, leaving little architecture-dependent signal for the contrast to detect.

The ordering of the three designs on this contrast, OL+BDC well above Hybrid and Hybrid above CO, is the substantive result. It is consistent with a single explanatory factor: the amount of uninterrupted off-drug time each design's structure exposes patients to. We note that this ordering is read at a common $N = 70$ and is not a statement about the designs' relative power (Section 4.1).

4.2 3.2 Mechanism: why the architectures diverge

Recall that Section 2.2.3 identified two channels through which carryover degrades the interaction signal, and only one of them is shared by both architectures. **Design-contrast erosion** shrinks $\text{Var}(D_{bc,it})$ as carryover pulls off-drug exposure values away from a clean 0/1 contrast. Because the interaction regressor is `bm:Dbc`, less contrast in `Dbc` means less information for the interaction coefficient under either architecture, and this channel alone explains mean moderation's modest, design-dependent loss (1.9-6.9%).

Conditional-variance inflation is unique to covariance moderation and is the channel that produces the much larger losses of Section 3.1. As $D_{bc,it} \to 0$ off drug, $\text{Var}(BR_{it} \mid B_i, D_{it}) = \sigma^2_{BR}(1 - c^2_{bm} D^2_{bc,it})$ rises from its on-drug floor of $\sigma^2_{BR}(1 - c^2_{bm})$ toward the full unconditional σ^2_{BR}. This converts variance that was explained by the biomarker into unexplained residual noise, which a homoscedastic fixed-effects model cannot separate from measurement error. Mean moderation has no such channel: its residual variance is constant by construction, so it loses only design contrast, never precision around a fixed contrast. This is why the population interaction coefficient stays essentially unbiased under both architectures (Section 3.3) while power diverges sharply. That is, the loss is a precision effect specific to covariance moderation, not a bias effect shared by both.

A rough, working-independence approximation to the interaction test's noncentrality parameter makes the connection to design and carryover explicit. Treating each observation's contribution to the Wald statistic as approximately additive and ignoring the `corCAR1` residual correlation that the actual analysis model estimates, the information available from an observation at exposure level $D_{bc,it} = x$, after partialling out the model's own `Dbc` main effect, is approximately proportional to $(x - \bar{D}_{bc})^2/\sigma_\epsilon^2$ under mean moderation and to $(x - \bar{D}_{bc})^2/[\sigma_{BR}^2(1 - c_{bm}^2 x^2)]$ under covariance moderation, where $\bar{D}_{bc}$ is the design's mean exposure level. Both summands shrink as carryover compresses $D_{bc,it}$ toward $\bar{D}_{bc}$ across the design, and the covariance-moderation summand is additionally discounted by the same conditional-variance-inflation factor derived above, so the total information lost accumulates fastest in designs that spend the most observation-weeks away from a clean on/off contrast.

We evaluated this approximation numerically against each of the three designs' exact timepoint and phase schedules (constructed with the package's own `buildtrialdesign()`, at $c_{bm} = 0.45$), rather than leaving it as an unexecuted next step, and the result is mixed. Summing $(D_{bc,it} - \bar{D}_{bc})^2$ over every observation in each design correctly orders the three by predicted information loss from $t_{1/2} = 0$ to $t_{1/2} = 1.0$: 36.8% in the focal Hybrid design, against 4.6% in CO and 51.2% in OL+BDC, matching the qualitative ordering of Section 3.1's power losses under either architecture. On the other hand, the covariance-moderation summand predicts an architecture-specific gap over mean moderation's own loss of under one percentage point at every design (37.1% in Hybrid, 4.7% in CO, 50.9% in OL+BDC) at $c_{bm} = 0.45$. That is far smaller than the large gap Section 3.1 actually reports (for example, Hybrid's 6.9% mean-moderation loss against its 15.4% covariance-moderation loss, or OL+BDC's 2.8% against 30.6%).

We should be clear that the exact conditional-variance-inflation identity itself is not in question here; it is independently confirmed against realized Monte Carlo variability in Section 3.3. What this numeric check shows instead is that the working-independence simplification used to connect that identity to design and carryover analytically discards enough of the `corCAR1` structure to badly underpredict the size of the architecture gap, and not only its near-absence in CO. A full analytical account of the magnitude, rather than only its direction, would need to incorporate the within-subject autocorrelation directly, and this remains undone. The qualitative account below is offered as the mechanism for CO's particular robustness in place of that missing derivation.

Qualitatively, the ordering of Section 3.1 follows the same information argument applied to each design's own pattern of drug-exposure alternation. Conditional-variance inflation needs sustained low-$D_{bc,it}$ time to accumulate. The longer a design lets $D_{bc,it}$ sit near zero before the next on-drug period resets it, the more of that period's information is converted to noise. OL+BDC sustains its off-drug window the longest of the three designs, so its variance-inflation channel does the most damage. CO alternates quickly enough, and leans on within-subject AR(1) correlation heavily enough to have already extracted most of the available signal by other means, that little conditional-variance inflation has room to act before the next on-drug period arrives. The Hybrid design's paths mix on- and off-drug periods at an intermediate cadence, so $D_{bc,it}$ rarely stays near zero as long as it does under OL+BDC. This caps how much variance inflation any single off-drug stretch can accumulate, and the design's intermediate architecture-dependent loss (Section 3.1) is the empirical signature of this intermediate exposure pattern.

4.3 3.3 Type I error and estimator bias

Across all 18 null cells of the Section 3.1 production run, Type I error sat in $[0.010, 0.074]$, with mean 0.042 and median 0.035, and no architecture-specific inflation. The uniformly sub-nominal rates are consistent with the `corCAR1` working-covariance miscalibration characterized in a companion calibration study[36], whose model-based standard error is over-stated by roughly 6 to 10 percent. Because this affects both architectures alike, it is common-mode with respect to the architecture contrast, and it does not affect it.

Mean estimated $\hat{\beta}_{bm:D_{bc}}$ at $c_{bm} = 0.45$ ranged from -0.230 to -0.281 across the 18 alternative cells, with the mean-moderation OL+BDC cell at $t_{1/2} = 1.0$ producing the largest absolute mean. Covariance-moderation cells, by contrast, maintained a mean near -0.234 across all $t_{1/2}$ values, exactly as the unbiasedness argument of Section 2.2.3 predicts. That is, the population interaction slope recovered from $D_{bc,it}$ does not depend on carryover once the analysis model's half-life matches the DGP's. Empirical SE of $\hat{\beta}_{bm:D_{bc}}$ runs 0.07 to 0.10 across cells, and the within-replicate model SE matches it at scale. This is consistent with nominal-95% Wald coverage, although it was not directly verified by a coverage simulation.

The conditional-variance-inflation mechanism of Section 3.2 can be checked directly against this same table, since it predicts where the empirical SE should grow with carryover and where it should not. We computed $|\hat{\beta}_{bm:D_{bc}}|/\mathrm{SE}(\hat{\beta}_{bm:D_{bc}})$ at $c_{bm} = 0.45$ as a design- and architecture-specific proxy for the interaction test's signal-to-noise ratio, and compared its value at $t_{1/2} = 0$ against $t_{1/2} = 1.0$. Under covariance moderation this ratio falls by 12.2% in OL+BDC (2.947 to 2.588) and 12.6% in Hybrid (3.259 to 2.849). Under mean moderation, on the other hand, it falls by only 5.4% in Hybrid (3.002 to 2.839) and is flat to slightly higher in OL+BDC (-2.7%, 2.865 to 2.943) and in CO (0.6%, mvn; 1.9%, mean moderation). This is the pattern Section 3.2 predicts. Covariance moderation's SNR erosion is driven almost entirely by SE growth (0.079 to 0.090 in OL+BDC, a 14.6% increase) against an essentially flat numerator, consistent with the derived $\sigma^2_{BR}(1 - c^2_{bm} D^2_{bc,it})$ variance-inflation identity. Mean moderation shows no comparable SE trend, because its residual variance does not depend on $D_{bc,it}$.

We verified this check directly from the committed cell-level summaries (`01-dgp-summary.txt`) rather than from a new simulation, and it substantiates the mechanism of Section 3.2 quantitatively. It remains, however, a post hoc check on realized Monte Carlo variability, rather than a derivation of the noncentrality parameter from design and DGP parameters alone.

5 4. Discussion

5.1 4.1 Matching architecture to biological mechanism

The results of Section 3 reduce to a single trade-off, which this subsection develops at length. On the one hand, mean moderation is simpler, is essentially immune to carryover, and is directly comparable to the overwhelming majority of the published biomarker-simulation literature. It assumes, however, a mechanism that is often biologically wrong, namely that the biomarker fixes each patient's drug response exactly, rather than merely predicting it. On the other hand, covariance moderation is biologically faithful in exactly the cases where mean moderation is not, because it lets the biomarker predict response only probabilistically. That fidelity is purchased at a real, design-dependent power cost under carryover, and it is a cost that an investigator who ran their power calculation under mean moderation alone would never see in their own simulation, and would therefore not budget for.

Neither architecture dominates the other in the abstract. The correct choice, and the size of the risk from choosing wrong, depends on the biology of the specific biomarker under study. This subsection substantiates that claim first from first principles and then from the finite-mixture formalism developed at length in a companion paper.

The choice of architecture is a choice about *why* the biomarker predicts response, and is therefore a biological statement rather than a coding preference. Mean moderation is appropriate when the biomarker mechanically sets the size of the drug effect, for example by fixing effective dose or receptor occupancy, so that doubling the biomarker roughly doubles the effect at every timepoint. Because carryover keeps the same proportion off drug, power is barely touched. Typical examples include drug clearance (kidney function setting the steady-state level of a renally cleared drug), receptor density, and metabolizer genotype.

Covariance moderation, by contrast, is appropriate when the biomarker is only a proxy for a hidden responder status that it indexes imperfectly, so that identical biomarker values can still respond differently, and the biomarker predicts response only while the drug is acting. Once the drug washes out, the correlation that carried the signal fades. The analysis model's point estimate stays close to unbiased even so (Section 2.2.3), but the precision with which it can be estimated drops, and power falls sharply as a result (Section 3.2). Typical examples include resting blood pressure as a marker of noradrenergic tone, inflammatory markers such as CRP or IL-6, and polygenic risk scores.

A fully faithful model of the hidden responder-class idea would be a finite mixture. Covariance moderation is, essentially, a convenient correlation-level approximation, adequate for power calculations but not a claim about the true mechanism. A companion paper[29] develops the mixture and latent-class formulations, deriving when a single multivariate normal reproduces a two-component mixture's first two moments and cataloguing where it fails. We note that Senn[14] also cautions that apparent biomarker-treatment effects are sometimes just variance artifacts, so covariance moderation's signal may be smaller than assumed.

The pmsimstats software was built for a prazosin/PTSD trial[25,37], in which the biomarker, resting blood pressure, is thought to be a proxy for noradrenergic tone rather than a direct driver of drug levels, a covariance-moderation story. The carryover-driven power loss we find under covariance moderation is therefore a clinically relevant warning: designs with long off-drug periods may lose more power to detect this biomarker than a mean-moderation simulation would suggest. Should blood pressure act through the mean channel as well as the correlation channel, the dual-channel architecture of the companion paper[30] is the appropriate model. Because it estimates two independent parameters, it calls for independent evidence on each.

Three further considerations support covariance moderation as the more defensible specification for this application. First, resting blood pressure is a single-occasion vital sign subject to substantial physiological and measurement variability, so treating it as a precise multiplicative scale factor on the treatment effect assumes a level of measurement precision the biomarker does not have. A specification that asserts only an association, and its strength, is proportionate to that measurement error.

Second, the distinction tracks an established one in the predictive-biomarker literature. The PATH framework of Kent et al.[21] separates a causal treatment effect modifier from a biomarker that is merely predictive of response without being causally upstream of it. Covariance moderation is the natural representation of the predictive role, and mean moderation of the causal-mediator role, so classifying blood pressure as predictive rather than causal in this trial argues for covariance moderation on grounds already established in that literature.

Third, mean moderation requires committing to a specific functional form for how the biomarker scales the treatment effect, typically linear in the centered biomarker. Covariance moderation makes the weaker assumption that biomarker and response are associated, without specifying the functional form of that association beyond its strength. This is the more parsimonious choice when investigators lack direct evidence for a particular dose-response shape.

The two architectures considered here are the two poles of a wider spectrum, and drug and biomarker properties can place a candidate application closer to one boundary of that spectrum than to either pole outright. A companion paper on latent-class and mixture-model formulations[29] develops this spectrum at length and grounds covariance moderation as a specific, derivable approximation to a genuinely latent-class generative mechanism. We summarize that argument here because it is the strongest available justification for covariance moderation's biological faithfulness and, at the same time, the source of its principal limitation.

The mixture argument begins from the observation that if a candidate biomarker indexes an unobserved responder or non-responder dichotomy, as the noradrenergic-tone hypothesis behind the prazosin and PTSD reference application supposes, the mechanistically faithful

DGP is not a correlation but a finite mixture,

$$Z_i \mid B_i \sim \text{Bernoulli}(\pi(B_i)), \qquad BR_{it} \mid Z_i = z \sim f_z(t, D_{bc,it}), \qquad \pi(B_i) = \text{expit}(\psi_0 + \psi_g\, b_i),$$

where Z_i is an unobserved class label fixed for participant i across the whole trial, b_i is the standardized biomarker, and the gating slope ψ_g governs how steeply the biomarker predicts class membership. Two properties of this generative form matter for what follows. First, Z_i is invariant across drug state and time: once a participant is drawn into class 1 or class 0, that membership persists through every on-drug and off-drug measurement occasion in the trial, in principle recoverable from the full within-subject trajectory even at timepoints where the instantaneous drug-exposure contrast has decayed to near zero. Second, the class-conditional mean trajectory itself still depends on $D_{bc,it}$, so a full carryover off-drug period drives the two class-conditional means toward each other if the responder and non-responder distinction lies purely in drug response, exactly the mean-blurring channel that Section 3.2 identifies as common to both architectures examined here.

Under within-class independence between B_i and BR_{it}, a two-component mixture of this form induces a marginal joint distribution of (B_i, BR_{it}) whose first two moments are reproduced, to leading order, by a single multivariate normal with $\text{Cor}(B, BR) = c_{bm}$, where

$$c_{bm}^2 = \frac{\pi(1-\pi)(\mu_B^1 - \mu_B^0)^2}{\text{Var}(B)} \cdot \frac{\pi(1-\pi)(\mu_{BR}^1 - \mu_{BR}^0)^2}{\text{Var}(BR)},$$

the product of the between-class variance fractions in the biomarker and in the biological-response component. Under mild regularity conditions (adequate class overlap, mixing proportion π bounded away from 0 and 1), this second-moment structure carries a differential correlation across drug-exposure conditions, elevated where the class-conditional means separate on drug and shrinking toward zero where they coincide off drug after carryover has run its course, which is precisely the covariance-moderation DGP of Section 2.2.2. Covariance moderation is therefore not an arbitrary alternative to mean moderation; it is the specific second-moment approximation to a latent-class mechanism that a biomarker acting as an imperfect proxy for a hidden responder subtype would actually generate, which is the formal grounding for treating it as the more biologically faithful choice whenever that subtype story is plausible.

That derivation also exposes covariance moderation's principal limitation directly. An MVN approximation reproduces only the first two moments of the mixture. It discards the class label Z_i itself, along with the class-invariance property that made Z_i carryover-robust in the true generative form. A correlation, unlike a class label, must be estimated fresh from the currently observed joint distribution at every timepoint, with no persistent per-participant memory once off-drug attenuation has taken effect. This is why the power losses documented in Section 3 are a property of the covariance-moderation *approximation* specifically, and not an inherent, unrecoverable cost of a genuinely latent-class biology. A class-aware analysis fitted to mixture-DGP data could, in principle, recover some of that apparent loss by using the carryover-invariant class label directly, up to an asymptotic bound set by the gap between the covariance-only and mixture-faithful power estimates. In practice, though, that recovery is bounded by identifiability: fitting a finite mixture (an EM algorithm, or its implementations in `lcmm` or `flexmix`) reliably distinguishes genuine class structure from noise only at sample sizes of several hundred to several thousand in the psychometric literature that developed these methods[38–40], well above the $N \in [30, 150]$ typical of N-of-1 biomarker-validation trials. At trial-relevant N, mixture analysis may be underpowered or unidentified even when the underlying biology is genuinely mixture-structured, so a class-aware primary analysis is not generally a usable alternative to the standard mixed-model analysis of Section 2.1, only a sensitivity check where identifiability permits. Covariance moderation is therefore best understood as the pragmatic compromise this identifiability constraint forces: the representation that is faithful to the underlying biology at the level of the joint distribution's first two moments, without requiring an estimation strategy that trial-relevant sample sizes cannot support. When the biomarker is a near-deterministic class indicator, for example a genotype with near-perfect sensitivity and specificity for a metabolizing versus non-metabolizing phenotype, the class-membership gradient is steep and a thresholded mean-moderation specification is the better approximation even though the underlying mechanism is a latent subtype rather than a graded scaling. When the two response classes are well separated, as in some targeted-oncology indications with a driver mutation that essentially abolishes response in its absence, the marginal response distribution is genuinely bimodal and neither architecture is adequate: a full mixture DGP, not examined by simulation in this paper, is required to avoid understating the power achievable by a class-aware analysis. When covariance, autocorrelation, or placebo-response parameters differ by latent class rather than only the mean or the biomarker correlation, a single-component MVN of either architecture is misspecified regardless of which moment carries the interaction. Investigators should therefore treat mean moderation and covariance moderation as convenient limiting cases to be checked against, rather than an exhaustive pair of options, whenever pilot data suggest the biomarker may index a genuinely discrete or highly separated subgroup structure.

The trial-design dependence documented in Section 3, largest in OL+BDC and smallest in CO, is the empirical signature of the same class-label discarding just established: because covariance moderation carries no persistent per-participant memory of responder status once off-drug attenuation has run its course, the signal it can offer a standard analysis shrinks with every additional unit of uninterrupted off-drug time a design imposes, in exactly the pattern Section 3.2 traces mechanistically across the three designs. So the design implication drawn in Section 4.4, that designs with long off-drug windows carry the largest architecture-dependent risk, should be read as a caution for trials analyzed with the standard mixed-model methods of Section 2.1 specifically, not as a statement that latent-class biology makes long off-drug windows safe by construction; a class-aware analysis, where trial-relevant identifiability permits one, is not bound by the same limitation. For the same reason, the ordering of the three designs on architecture-sensitivity (Section 3.1) is a statement about how much power a fixed analysis model loses to the architecture choice, not a design-preference recommendation. At matched $N = 70$, covariance-moderation power under carryover happens to land close together for Hybrid (0.711) and CO (0.723) and well below both for OL+BDC (0.539), but the paths are allocated unevenly across the designs (18/18/17/17 for Hybrid's four paths versus 35/35 for CO's and OL+BDC's two), and design selection in practice turns on feasibility, blinding, and regulatory considerations that this power comparison alone does not capture. Investigators choosing among designs on carryover-robustness grounds should treat the Section 3.1 ordering as a hypothesis to examine under their own design's path allocation, not as a ranking to adopt directly.

5.2 4.2 Reporting recommendations

Criterion	Mean moderation	Covariance moderation
Biomarker role	Causal mediator	Statistical predictor
Mechanism	Deterministic scaling	Probabilistic association
Signal location	Mean structure	Covariance structure
Carryover impact	Modest (up to 7%)	Design-dependent, up to 31%
Appropriate when	Biomarker determines dose or PK	Biomarker indexes a latent subtype

The choice of DGP architecture should be guided by the hypothesized biological mechanism, as summarized above. When the mechanism is ambiguous, as it often is in early biomarker development, we recommend that investigators run the simulation under both architectures and report the range of power estimates. We suggest treating covariance moderation as the conservative default, because its larger carryover sensitivity keeps power estimates from being optimistic. A dual-channel sensitivity sweep across mixing weights is available in the companion paper[30]. Designing the trial to minimize carryover, through adequate washout periods, also reduces sensitivity to the architecture choice. More generally, simulation studies evaluating biomarker-moderated treatment effects should state explicitly the following.

- Whether the interaction is generated through mean moderation, covariance moderation, or both channels simultaneously.
- The biological rationale for the chosen mechanism.
- Whether any carryover model operates on the interaction signal consistently with that choice.
- When the mechanism is uncertain, sensitivity analyses under the alternative architecture.

These requirements specialize the Data-generating element of the ADEMP framework[33] to the class of architecture-sensitive biomarker simulations.

5.3 4.3 Possible mitigation strategies

Under covariance moderation, the power loss has two sources. The drug-exposure variable's variance shrinks, which is recoverable, and the biomarker-response correlation genuinely fades off drug, which is lost information no model can recover. Because off-drug points carry less signal, methods that down-weight or drop them may recover part of the loss. Candidates include restricting the analysis to on-drug points, weighting observations by expected residual drug effect (via the `weights` argument in `nlme::lme`), a within-subject contrast regressing each patient's on-minus-off mean difference on the biomarker, a two-stage random-slopes approach, or excluding the most contaminated early off-drug points. We have not tested any of these strategies by simulation here; a companion paper on carryover-mitigation strategies takes that up. We regard a systematic simulation comparison, analogous to the factorial comparison of Section 3, as important future work, and we expect the on-drug-only restriction and the weighted analysis to be the most tractable candidates for a first empirical evaluation.

5.4 4.4 Relation to the wider literature

Almost every trial-simulation framework uses mean moderation. Simon[22], Freidlin and Korn[23], Renfro and Sargent[41], and standard N-of-1 simulation studies all put the biomarker effect in the mean. We note that even methodological work on N-of-1 carryover sharing authors with Hendrickson et al.[25] uses mean moderation, which suggests Hendrickson's differential-correlation choice was a one-off rather than a convention. The practical worry is that most published power estimates may be optimistic for biomarkers that really behave like covariance moderation, because they ignore the extra carryover-driven loss demonstrated here.

The distinction matters most exactly in the designs precision medicine favors. Treatment-switching designs (crossover, N-of-1 hybrid, basket/platform trials) put patients through off-drug phases where the covariance-moderation correlation decays, so the interaction signal in later periods is weaker than in the first. Parallel-group enrichment designs, with no off-drug phase, are immune, and give similar power under either architecture. Adaptive enrichment trials that use interim power calculations written in the mean-moderation idiom are at particular risk: if the truth is closer to covariance moderation and patients pass through washouts, the real power is lower than the rule assumes, which can trigger premature stopping or an over-narrow eligible population. This is a data-generating-process question, separate from, and not answered by, the analysis-model guidance in the PATH statement[21]. In sum, the architecture choice matters most in exactly the within-subject designs precision medicine finds most attractive, and least in the parallel-group designs where it is often, wrongly, assumed to be harmless.

5.5 4.5 Limitations

Several limitations qualify these results. The magnitude of the covariance-moderation loss in the focal Hybrid design (15.4%), and the largest value observed across the three designs, the 30.6% loss in OL+BDC, both fall below the 40-60% range suggested by earlier exploratory runs in this project. The precise figures should therefore be read as design- and calibration-specific rather than universal. A confirmatory run at $N = 140$ on the Hybrid design, analogous to the $N = 140$ OL+BDC check reported in an earlier version of this analysis, has not yet been re-executed under the current N-matching convention (see Reproducibility) and remains future work.

The uniformly sub-nominal Type I error we observe reflects a known conservatism of the `corCAR1` Wald test that is common-mode across architectures but not itself resolved here. A cluster-robust remedy is characterized in a companion calibration study[36]. We note that nominal coverage was inferred from the correspondence between empirical and model-based standard errors, rather than verified by a dedicated coverage simulation. The mitigation strategies of Section 4.3 are mechanistic hypotheses, not simulation-tested recommendations. Finally, the combined architecture that activates both channels at once is deferred entirely to a companion paper[30], and our results are matched to a single reference application (prazosin/PTSD) and three within-subject designs. Generalization to other biomarkers, designs, or effect sizes should be checked rather than assumed.

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6 5. Conclusions

1. The choice between direct mean moderation and covariance moderation for generating biomarker-treatment interactions in clinical trial simulations has substantive consequences for power estimates under carryover.

2. Covariance moderation has intuitive appeal whenever a biomarker is better understood as predictive than causal, and Section 4.1 sets out the reasons this is often the more defensible specification. That appeal carries a measurable cost under carryover, and the cost is design-dependent, because it is driven by a mechanism (Section 3.2) that is itself design-dependent: carryover erodes the biomarker-response correlation that carries the covariance-moderation signal only during uninterrupted off-drug time, so the loss scales with how long each design sustains an off-drug period. In the focal Hybrid design, carryover reduces power by 15.4% under covariance moderation against 6.9% under mean moderation, roughly double. The crossover and open-label-plus-blinded-discontinuation contrast designs bracket this from opposite sides, from 3.0% (crossover, not distinguishable from zero, because within-subject AR(1) correlation already carries most of the signal) to 30.6% (OL+BDC, whose sustained blinded off-drug phase gives the correlation-erosion mechanism the most uninterrupted time to act).

3. The choice should be guided by the hypothesized biological mechanism. Mean moderation is appropriate for biomarkers that directly determine drug pharmacokinetics or pharmacodynamics, covariance moderation for biomarkers that statistically predict treatment response through latent subtype membership.

4. For the prazosin-PTSD application, with blood pressure as the predictive biomarker, covariance moderation is the more appropriate model, and the resulting power sensitivity to carryover should inform trial-design decisions about off-drug period duration.

5. The existing clinical trial simulation literature uses mean moderation almost exclusively. The Hendrickson et al. (2020) MVN approach is, to the best of our knowledge, unique in the N-of-1 context. This means published power estimates for biomarker-moderated crossover and N-of-1 designs may be systematically optimistic for biomarkers that operate through a correlation-based mechanism.

6. Simulation studies for biomarker-moderated treatment effects should explicitly report their DGP architecture and, when the biological mechanism is uncertain, consider sensitivity analyses under the alternative.

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7 Reproducibility

N denotes the total number of participants across all randomization paths, and Section 3.1 runs every design at the same total, $N = 70$, allocated across that design's paths. The cross-design comparison in Section 3 is therefore a design comparison rather than a sample-size comparison. We note that an earlier version of this analysis fixed the per-path count at 35 instead of the total, which gave the four-path Hybrid design 140 participants against 70 for the two-path designs and confounded any Hybrid advantage with twice the sample. Section 3.1 has since been re-run at matched totals, and only the Hybrid figures changed. A confirmatory run at $N = 140$ on the Hybrid design, to check that the architecture-dependent ordering survives the power saturation that comes with a larger sample, is planned but has not yet been executed under the current N-matching convention, and is not reported here.

The Hendrickson et al. comparison arm of Section 2.2.4, including the vendored original source (with its one documented compatibility patch) and the positive-definiteness sweep, is at `analysis/scripts/quick-sim/hendrickson-original-comparison/`, with its output at `analysis/data/quick-sim/hendrickson-original-comparison/`. A structured, section-by-section summary of the argument, without the derivations, tables, or citations, is provided as a supplementary document (`bullets.Rmd`).

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9 Appendix A. Why mean and covariance moderation

9.1 A.1 The two channels exhaust the Gaussian case

The simulations in this paper draw participant trajectories from a multivariate normal distribution, and a multivariate normal is completely determined by its mean vector μ and its covariance matrix Σ. Any biomarker-treatment interaction written into such a data-generating process must therefore enter through μ, through Σ, or through both. Mean moderation is the first route, covariance moderation the second, and the combined architecture developed in a companion paper[30] is the third. Within the Gaussian family there is no fourth place to put the interaction. This is why the two architectures are treated as a pair rather than as two entries in an open-ended list: they are not two options among many, but the two components of a decomposition.

The exhaustiveness claim is specific to the Gaussian family and should not be read more broadly. A data-generating process built on a finite mixture, or on explicit latent responder classes, can carry an interaction in features that neither μ nor Σ captures for a single component, and Senn notes finite mixtures as an alternative variance-component model[17]. Covariance moderation is best understood as the second-moment reduced form of such a mixture, which is why the latent-class treatment is developed separately[29].

9.2 A.2 What each channel asserts

The two channels are not merely different parameterizations of one mechanism; each commits to a distinct claim about what the biomarker is.

Mean moderation asserts that the biomarker fixes a participant's drug effect. Two participants sharing a biomarker value share an expected treatment effect exactly, and any difference between their realized responses is measurement error. This is the appropriate commitment when the biomarker measures pathway engagement directly, or stands in a causal path between treatment and response.

Covariance moderation asserts that the biomarker predicts response only probabilistically. Two participants sharing a biomarker value may still respond differently, because the biomarker indexes an unobserved responder status imperfectly rather than determining response. This is the appropriate commitment when the biomarker is a statistical proxy, which is the situation the prazosin and PTSD application was designed around.

The distinction is substantive rather than technical. It determines what a fitted interaction coefficient means, and, as Section 3 shows, how much power carryover costs.

9.3 A.3 Standing in the literature

Mean moderation is the prevailing convention wherever biomarker-treatment interactions are simulated. Haller and Ulm[24] place the interaction in a regression mean, as do the oncology design literatures on enrichment and predictive-marker trials[22,23]. Grenet and colleagues[31] use the same form for a crossover. That literature is, however, overwhelmingly parallel-group.

The N-of-1 methodological literature does not adopt mean moderation, because it does not model an observed moderator at all. Its standard formulation carries a random participant-by-treatment interaction, a variance component describing unexplained heterogeneity with no covariate attached[14,16,17]. Biomarker-validation simulation for N-of-1 designs is therefore close to unoccupied territory rather than a settled field with a convention to follow, and Hendrickson et al.[25] is its principal entry. The present paper adopts the two-channel decomposition of A.1 precisely because no established N-of-1 convention exists to inherit.

9.4 A.4 Holding the analysis model fixed

Both architectures are analyzed with the identical model of Section 2.1. This is deliberate and worth stating explicitly, because a data-generating process and an analysis model are separate objects, and a comparison that varies both attributes differences to neither. The continuous exposure variable `Dbc` loses contrast variance as carryover grows, and a binary on-drug indicator does not. A comparison in which one architecture carried a continuous regressor and the other a binary one would confound that effect with the architecture contrast it was meant to isolate. Holding the analysis model fixed across architectures means that every difference reported in Section 3 is attributable to the data-generating process alone.

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10 Appendix B. Correlation structure of the covariance-moderation and published data-generating processes

This appendix sets out the correlation matrix that each covariance-based data-generating process assembles, so that the differences discussed in Section 2.2.4 can be read off directly. Throughout, `covar` denotes Architecture B as defined in Section 2.2.2 and `orig` denotes the published implementation at commit `58b32a9`. Symbols follow the variable names used in the code: BM is the biomarker, written B_i in Section 2.1, and BR_p the biological response component at occasion p, written BR_{it} there. Mean moderation does not appear below, because it writes no biomarker-response correlation into the covariance at all; its vector b is identically zero and its interaction is carried in the mean instead.

10.1 B.1 Variable ordering and block partition

Both implementations draw each participant's trajectory from a single multivariate normal over the same ordered variable list. With n measurement occasions at cumulative weeks $w_1 < w_2 < \cdots < w_n$, the vector is

$$X = \big(\,\mathrm{BM},\ \mathrm{BL},\ TV_1,\dots,TV_n,\ PB_1,\dots,PB_n,\ BR_1,\dots,BR_n\,\big)^{\top}, \qquad \dim X = 2+3n.$$

The ordering is factor-major: all n occasions of one factor are contiguous, rather than all three factors at one occasion. Writing A_c for the $n \times n$ within-factor block of factor $c \in \{tv, pb, br\}$ and $C_{cc'}$ for the $n \times n$ cross-factor block, the correlation matrix partitions as

$$R \;=\; \begin{pmatrix} 1 & 0 & 0^{\top} & 0^{\top} & b^{\top} \\ 0 & 1 & 0^{\top} & 0^{\top} & 0^{\top} \\ 0 & 0 & A_{tv} & C_{tv,pb} & C_{tv,br} \\ 0 & 0 & C_{pb,tv} & A_{pb} & C_{pb,br} \\ b & 0 & C_{br,tv} & C_{br,pb} & A_{br} \end{pmatrix}.$$

Two features hold in both implementations. The baseline term BL is uncorrelated with every other variable, so its row and column are those of the identity. The biomarker BM couples only to the response factor BR, through the vector $b \in \mathbb{R}^n$; its correlations with TV and PB are zero. The two implementations differ in the interior of A_c, in the interior of $C_{cc'}$, and in b.

10.2 B.2 Within-factor blocks A_c

Let ρ_c denote the autocorrelation parameter of factor c (`c.tv`, `c.pb`, `c.br` in the code).

10.2.0.1 `orig`: compound symmetry. A single constant is assigned to every pair of occasions, irrespective of their separation:

$$[A_c]_{ij} = \begin{cases} 1, & i = j \\ \rho_c, & i \neq j \end{cases} \qquad \Longleftrightarrow \qquad A_c = (1-\rho_c) I_n + \rho_c J_n,$$

with J_n the $n \times n$ matrix of ones. For $n = 4$,

$$A_c^{\mathrm{orig}} = \begin{pmatrix} 1 & \rho_c & \rho_c & \rho_c \\ \rho_c & 1 & \rho_c & \rho_c \\ \rho_c & \rho_c & 1 & \rho_c \\ \rho_c & \rho_c & \rho_c & 1 \end{pmatrix}.$$

The block has two distinct eigenvalues, $1+(n-1)\rho_c$ once and $1-\rho_c$ with multiplicity $n-1$, so positive definiteness requires $-1/(n-1) < \rho_c < 1$. The upper bound on admissible correlation therefore tightens as occasions are added.

10.2.0.2 `covar`: AR(1) in calendar time. The correlation decays with the elapsed time between occasions, using the cumulative week gap rather than the index difference:

$$[A_c]_{ij} = \rho_c^{|w_i - w_j|}.$$

For $n = 4$, writing $d_{ij} = |w_i - w_j|$,

$$A_c^{\text{covar}} = \begin{pmatrix} 1 & \rho_c^{d_{12}} & \rho_c^{d_{13}} & \rho_c^{d_{14}} \\ \rho_c^{d_{12}} & 1 & \rho_c^{d_{23}} & \rho_c^{d_{24}} \\ \rho_c^{d_{13}} & \rho_c^{d_{23}} & 1 & \rho_c^{d_{34}} \\ \rho_c^{d_{14}} & \rho_c^{d_{24}} & \rho_c^{d_{34}} & 1 \end{pmatrix}.$$

Because the exponent uses calendar time, unequally spaced designs produce unequal decay across consecutive occasions. In the Hybrid design, whose occasions fall at weeks $4, 8, 9, 10, 11, 12, 16, 20$, consecutive gaps of one week during the discontinuation phase yield much higher adjacent correlations than the four-week gaps at either end.

10.3 B.3 Cross-factor blocks $C_{cc'}$

Let c_1 denote the same-occasion cross-factor correlation (`c.cf1t`) and $c_\times$ the different-occasion cross-factor correlation (`c.cfct`).

10.3.0.1 `orig`. Both are constants:

$$[C_{cc'}]_{ij} = \begin{cases} c_1, & i = j \\ c_\times, & i \neq j \end{cases} \qquad \Longleftrightarrow \qquad C_{cc'} = c_\times J_n + (c_1 - c_\times) I_n.$$

10.3.0.2 `covar`. The same-occasion entry is unchanged, but the different-occasion entries inherit the AR(1) decay:

$$[C_{cc'}]_{ij} = \begin{cases} c_1, & i = j \\ c_\times\, \rho^{|w_i - w_j|}, & i \neq j. \end{cases}$$

10.4 B.4 Biomarker coupling vector b

The vector b carries the biomarker-treatment interaction. This is the only place in R where the two implementations differ in mechanism rather than in decay profile.

10.4.0.1 `covar`.

The entry is gated on the drug indicator and decays with time since discontinuation $t_{sd,p}$ at rate $\lambda = \ln 2/t_{1/2}$:

$$b_p = \begin{cases} c_{bm}, & \text{occasion } p \text{ on drug} \\ c_{bm}\, e^{-\lambda t_{sd,p}}, & \text{occasion } p \text{ off drug.} \end{cases}$$

10.4.0.2 `orig`.

The entry is gated on whether the response mean $\mu_{BR,p}$ is nonzero, and takes only two values:

$$b_p = \begin{cases} c_{bm}, & \mu_{BR,p} \neq 0 \\ 0, & \mu_{BR,p} = 0. \end{cases}$$

Because carryover keeps $\mu_{BR,p}$ nonzero after discontinuation, the position of this step depends on the carryover half-life. At $t_{1/2} = 0$ the step coincides with the drug indicator and the two vectors are equal entry for entry. As $t_{1/2}$ grows they separate in opposite directions: the `covar` entries decay toward zero while the `orig` entries rise to c_{bm}. At $t_{1/2} = 1.0$ in the Hybrid design, $b^{\text{orig}} = c_{bm}\mathbf{1}_n$, so the vector carries no on-drug versus off-drug contrast at all.

10.5 B.5 A worked example

To make the three components concrete, take four occasions drawn from the Hybrid design at cumulative weeks $w = (10, 11, 12, 16)$, with the first and fourth on drug and the second and third off it, so that $t_{sd} = (0, 1, 2, 0)$. Set $\rho_{br} = 0.6$, $\rho_{pb} = 0.4$, $c_1 = 0.3$, $c_\times = 0.1$, $c_{bm} = 0.45$, and a carryover half-life of one week, giving $\lambda = \ln 2 \approx 0.693$. The matrix of week gaps is

$$(|w_i - w_j|) = \begin{pmatrix} 0 & 1 & 2 & 6 \\ 1 & 0 & 1 & 5 \\ 2 & 1 & 0 & 4 \\ 6 & 5 & 4 & 0 \end{pmatrix}.$$

The unequal spacing matters: occasions one to three are a week apart, while the fourth sits four weeks after the third.

10.5.1 B.5.1 Within-factor block A_{br}

$$A_{br}^{\text{orig}} = \begin{pmatrix} 1.0000 & 0.6000 & 0.6000 & 0.6000 \\ 0.6000 & 1.0000 & 0.6000 & 0.6000 \\ 0.6000 & 0.6000 & 1.0000 & 0.6000 \\ 0.6000 & 0.6000 & 0.6000 & 1.0000 \end{pmatrix}, \qquad A_{br}^{\text{covar}} = \begin{pmatrix} 1.0000 & 0.6000 & 0.3600 & 0.0467 \\ 0.6000 & 1.0000 & 0.6000 & 0.0778 \\ 0.3600 & 0.6000 & 1.0000 & 0.1296 \\ 0.0467 & 0.0778 & 0.1296 & 1.0000 \end{pmatrix}.$$

Under `orig` every pair of occasions carries 0.6, so the week-apart pair $(1, 2)$ and the six-weeks-apart pair $(1, 4)$ are treated as equally associated. Under `covar` the same two pairs carry 0.6 and 0.0467, a factor of thirteen apart. The consequence for the fourth occasion is the striking one: it is nearly independent of the first three under AR(1), and as strongly tied to them as they are to each other under compound symmetry.

10.5.2 B.5.2 Cross-factor block $C_{tv,pb}$

$$C^{\text{orig}}_{tv,pb} = \begin{pmatrix} 0.3000 & 0.1000 & 0.1000 & 0.1000 \\ 0.1000 & 0.3000 & 0.1000 & 0.1000 \\ 0.1000 & 0.1000 & 0.3000 & 0.1000 \\ 0.1000 & 0.1000 & 0.1000 & 0.3000 \end{pmatrix}, \qquad C^{\text{covar}}_{tv,pb} = \begin{pmatrix} 0.3000 & 0.0400 & 0.0160 & 0.0004 \\ 0.0400 & 0.3000 & 0.0400 & 0.0010 \\ 0.0160 & 0.0400 & 0.3000 & 0.0026 \\ 0.0004 & 0.0010 & 0.0026 & 0.3000 \end{pmatrix}.$$

The diagonal, which carries the same-occasion cross-factor correlation c_1, is identical in both. The off-diagonal entries decay under `covar` and do not under `orig`, and by the fourth occasion the difference is two orders of magnitude.

A caution attaches to the `covar` block. In the implementation the decay rate applied to a cross-factor pair is taken from whichever factor the outer loop is currently visiting, and because each unordered pair is visited in both orders, the later write survives. The rate appearing in $C_{tv,pb}$ is therefore determined by loop order rather than by any modeling decision. The values above use ρ_{pb} on that basis. This is an implementation artifact and not a property of the AR(1) specification.

10.5.3 B.5.3 Biomarker vector b

At a carryover half-life of one week,

$$b^{\text{covar}} = (0.4500,\, 0.2250,\, 0.1125,\, 0.4500)^\top, \qquad b^{\text{orig}} = (0.4500,\, 0.4500,\, 0.4500,\, 0.4500)^\top.$$

The two off-drug entries halve and quarter under `covar`, tracking $e^{-\lambda t_{sd}}$, whereas under `orig` they take the full c_{bm}, because carryover has left the response mean nonzero and the step therefore never fires.

Removing carryover collapses the difference entirely:

$$b^{\text{covar}} = b^{\text{orig}} = (0.4500,\, 0.0000,\, 0.0000,\, 0.4500)^\top \qquad (t_{1/2} = 0).$$

This pair of displays is the essential comparison. The vector b is the only channel through which the biomarker-treatment interaction enters the covariance, and an interaction is identified by b varying between on-drug and off-drug occasions. At $t_{1/2} = 0$ the two implementations produce the same vector. At any positive half-life b^{orig} is constant, so it carries no on-drug versus off-drug contrast, while b^{covar} retains one. Across the grid $t_{1/2} \in \{0, 0.5, 1.0\}$ used in this paper, `orig` therefore supplies one cell identical to `covar` and two in which its interaction channel is degenerate.

10.6 B.6 The three data-generating processes

Organized by where each process writes the biomarker-treatment interaction into the multivariate normal (μ, Σ). All three share the same variable ordering and block partition of Section B.1, and differ only in the entries listed.

	`mean`	`covar`	`orig`
Interaction written into	μ	Σ	Σ
Interaction channel	BR mean shift	b vector	b vector
A_c within-factor	AR(1) $\rho_c^{\lvert w_i - w_j \rvert}$	AR(1) $\rho_c^{\lvert w_i - w_j \rvert}$	compound symmetry, constant ρ_c
$C_{cc'}$ diagonal	c_1	c_1	c_1
$C_{cc'}$ off-diagonal	$c_\times \rho^{\lvert w_i - w_j \rvert}$	$c_\times \rho^{\lvert w_i - w_j \rvert}$	constant $c_\times$
b on drug	0	c_{bm}	c_{bm}
b off drug	0	$c_{bm} e^{-\lambda t_{sd}}$	0 or c_{bm}
b off-drug profile	n/a	graded	step
b gate	n/a	`onDrug`, then t_{sd}	$\mu_{BR} \neq 0$
Gate position carryover-dependent	n/a	no	**yes**
μ shift on drug	$c_{bm} z_i \sigma_{BR}$	none	none
μ shift off drug	0 (step)	none	none
Carryover on BR mean	anchored: $\mu_{BR,p^*}(1/2)^{t_{sd}/t_{1/2}}$		recursive: $\mu_{BR,p-1}(1/2)^{t_{sd}/t_{1/2}}$
BL row and column	identity in all three		
BM–TV, BM–PB	zero in all three		

Three features of this comparison are worth drawing out.

First, `mean` and `covar` share their entire covariance matrix apart from the vector b. Setting b to zero and adding the mean shift converts one into the other, with every within-factor and cross-factor block untouched. This is the decomposition of Appendix A made concrete: the two architectures are not different models but the two places a single joint distribution can carry an interaction.

Second, `orig` differs from `covar` on three rows simultaneously, in A_c, in the off-diagonal of $C_{cc'}$, and in b. No single row can therefore be held responsible for any difference in results obtained under it, and a comparison against `orig` bounds the joint effect of the three differences rather than decomposing them. Separating them would require running the correlation structure and the correlation gate as crossed factors.

Third, only the b row differs in mechanism rather than in decay profile. The A_c and $C_{cc'}$ rows describe the same quantity decaying at different rates, or not decaying at all. The b row describes different conditions being tested to decide whether the interaction is present. That distinction is what makes the gate row consequential: because `orig` tests the response mean rather than the drug indicator, and carryover keeps the response mean nonzero, its gate opens at off-drug occasions precisely when carryover is present, and its interaction channel loses the on-drug versus off-drug contrast on which identification depends.

The final two rows record what all three processes hold in common. The baseline term is uncorrelated with everything, and the biomarker couples to no factor other than the response.

The carryover row requires explanation, since the three processes agreed on it until recently and a reader comparing code will want to identify the difference quickly.

All three adjust the response mean at an off-drug occasion by adding a decayed fraction of the previous response. They differ in what is decayed. Writing p^* for the most recent on-drug occasion,

$$\underbrace{\mu_{BR,p} + \mu_{BR,p^*}\,(1/2)^{t_{sd,p}/t_{1/2}}}_{\texttt{mean, covar}} \qquad \text{against} \qquad \underbrace{\mu_{BR,p} + \mu_{BR,p-1}\,(1/2)^{t_{sd,p}/t_{1/2}}}_{\texttt{orig}}\,.$$

The published form multiplies the *already adjusted* mean at $p-1$ while $t_{sd,p}$ is measured cumulatively from discontinuation, so elapsed time is counted twice. The corrected form applies the cumulative factor once, to the mean at discontinuation.

Two consequences make the difference easy to spot in output. The first off-drug occasion of any run agrees exactly under both forms, since there is nothing yet to inherit. Divergence begins at the second. In the worked example of Section B.5, at $t_{1/2} = 1$ week, occasion 3 carries $0.25\,\mu_{BR,1}$ under the corrected form and $0.125\,\mu_{BR,1}$ under the published one, a factor of two. The gap widens with each further off-drug occasion, so designs with long uninterrupted washouts show it most.

We corrected `mean` and `covar` in September 2026 and leave `orig` as published, since the object of that column is to describe the published code rather than to improve it.

The consequence for the re-run flagged in Section 2.2.4 should be settled before that run is designed. A comparison of `covar` against `orig` now carries one difference beyond the four enumerated there, and it is of our making. Running `covar` as corrected asks what the architecture delivers today and credits the correction to `covar`. Running `covar` with the recursion restored isolates the four architectural differences and answers the question Section 2.2.4 poses. We prefer the second for the headline comparison, with the first reported alongside it, so that the cost of the correction remains visible.

10.7 B.7 Summary of differences

Component	`orig`	`covar`
A_c interior	ρ_c	$\rho_c^{\lvert w_i - w_j \rvert}$
$C_{cc'}$, $i = j$	c_1	c_1
$C_{cc'}$, $i \neq j$	$c_\times$	$c_\times \rho^{\lvert w_i - w_j \rvert}$
b_p, on drug	c_{bm}	c_{bm}
b_p, off drug	0 or c_{bm}	$c_{bm} e^{-\lambda t_{sd,p}}$
BL row/column	identity	identity
BM–TV, BM–PB	0	0

The compound-symmetry form constrains ρ_c more tightly than the AR(1) form as n grows, which is the mechanism behind the narrower positive-definite range of c_{bm} under `orig`.